AMERICAN UNIVERSITY OF BEIRUT
MAROUN SEMAAN FACULTY OF ENGINEERING & ARCHITECTURE

Department of Industrial Engineering and Management (IEM)

Rate of Return

Engineering Economy

What is the Rate of Return?

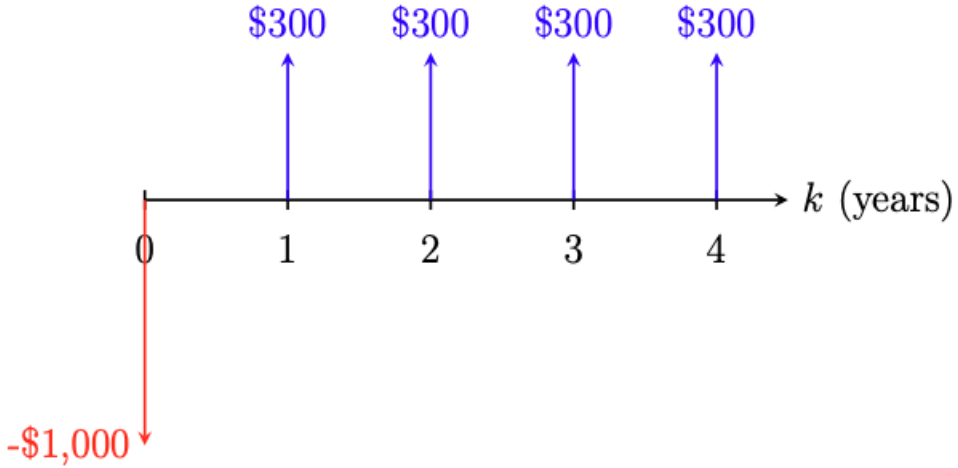
And why is it a convenient measure?



ROR - Definition

$i^*: PW(i^*) = 0$ where $i^* \in [-1, \infty)$

- The rate of return i^* is the interest rate that sets the present worth of all cash flows to zero.

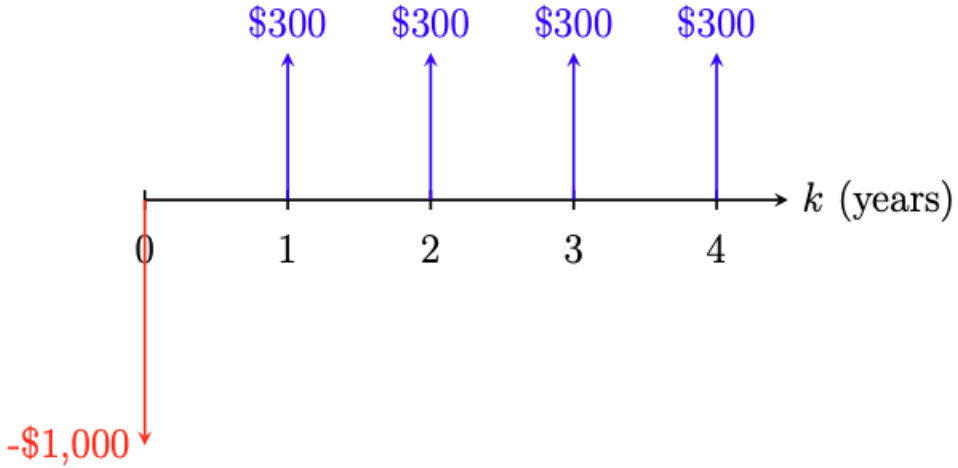


- ROR ≠ Payback Period (Ch5)

ROR - Definition

$i^*: PW(i^*) = 0$ where $i^* \in [-1, \infty)$

- The rate of return i^* is the interest rate that sets the present worth of all cash flows to zero



ROR is the **rate of interest i^*** paid on the **unrecovered balance of an investment**, so that the final receipt brings the **balance to exactly zero with interest considerations**.

Microsoft Excel Worksheet

ROR vs Interest Rates (MARR)

Interest Rate or MARR

- **External** Benchmark: set by the market
- **Input** to our cashflow modification
- What cashflows **could** earn if we invest it risk free

Rate of Return (ROR)

- Internal property of the cash flow
- **Output** from our cashflow
- What our money is **actually** earning

Decision Criteria and Benefits

For a Single Alternative

Decision Criteria – Single Alternative

- $i^* \geq MARR$ → **Accept** alternative (Economically not viable)
- $i^* < MARR$ → **Reject** alternative (Economically viable)

WHY?

Pros and Cons

Pros

- Knowledge of the MARR is not required
- Percentage format is intuitive

Cons

- Difficult to solve for complex cash flows
- Sometimes we have multiple i^*

Finding i^*

For a Single Alternative



Calculation – PW

Solve $PW(i^*) = 0$

$$PW_{\boldsymbol{D}isbursment} = PW_{\boldsymbol{R}eceipts}$$

$$\sum_{t=0}^{n} F_t(1+i)^{-t} = 0$$



Solution Procedure

1. Convert all disbursement into EITHER a single OR uniform amounts **without** consider the time value of money
2. Convert all receipt into either a single or uniform amounts
3. Solve $PW(i) = 0 \rightarrow PW_D = PW_R$ to get an *initial guess*
4. Refine and interpolate

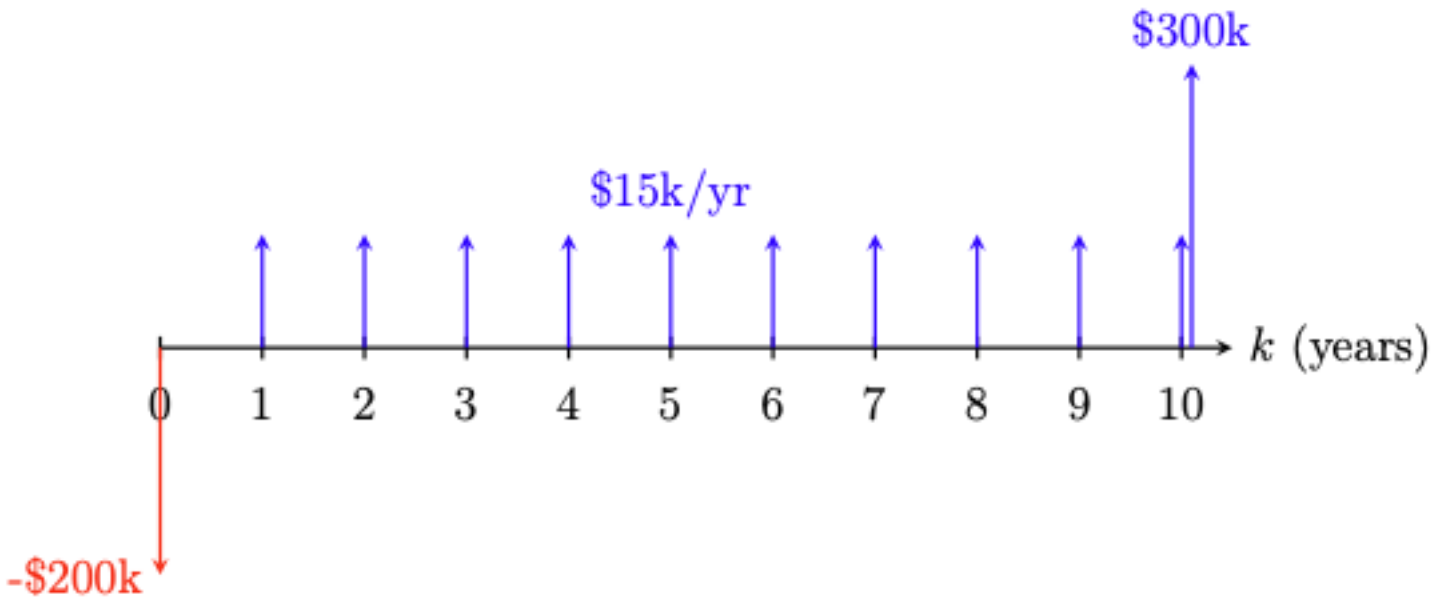
We need one factor only to find an initial guess



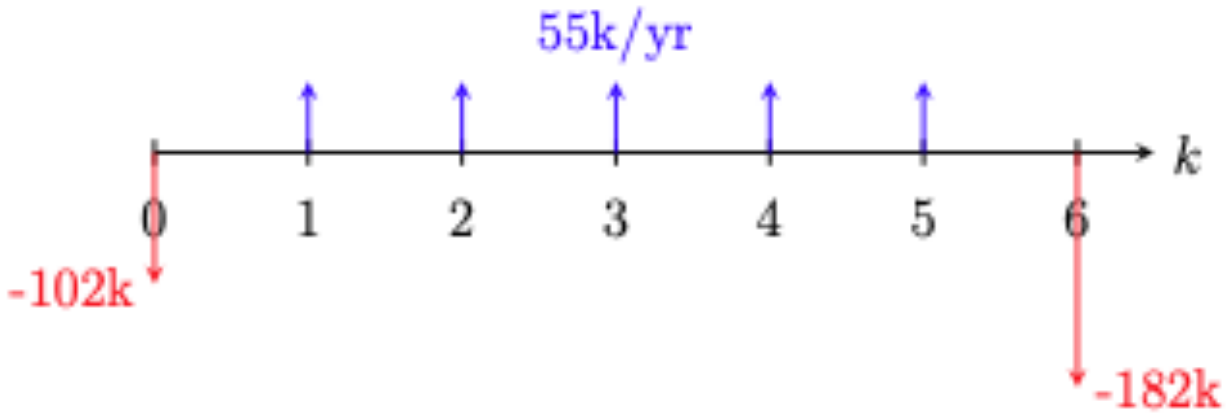
Compute i^* - Example 1

Engineers at Colorful Paints recommend an investment of $200,000 now in sustainable manufacturing methods. This investment will reduce waste and generate estimated savings of $15,000 per year for the next 10 years. Additionally, there will be a one-time savings of $300,000 at the end of 10 years in facility upgrade costs. Determine the rate of return (i^*) for this investment using manual estimation.

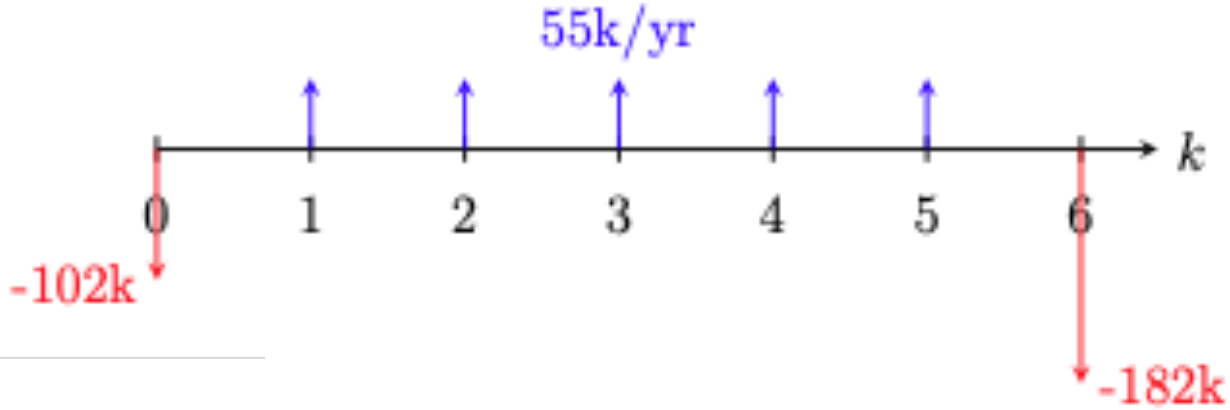
Microsoft Excel Worksheet

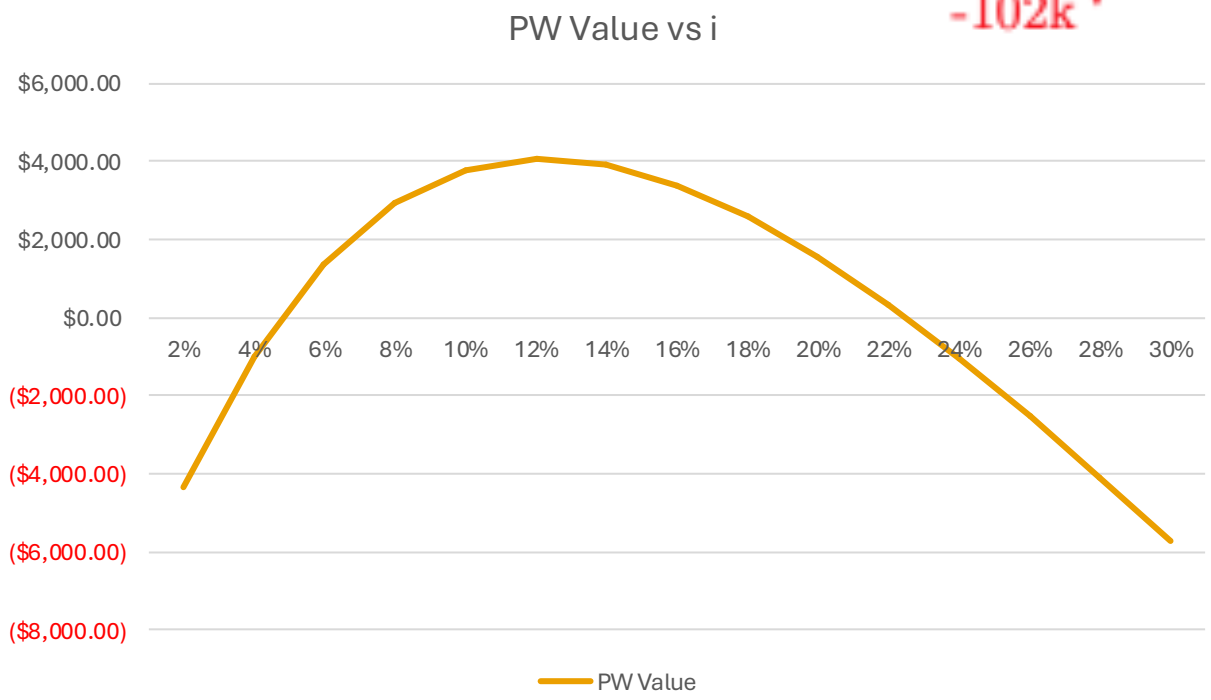

$300k
$15k/yr
k (years)
0
1
2
3
4
5
6
7
8
9
10
-$200k

Compute i^* - Example 2



Compute i^* - Example 2





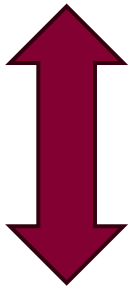
Multiple RoR



Identifying Multiple ROR Values

Descartes' Rule

Number of sign changes in the cash flows



Maximum number of ROR values

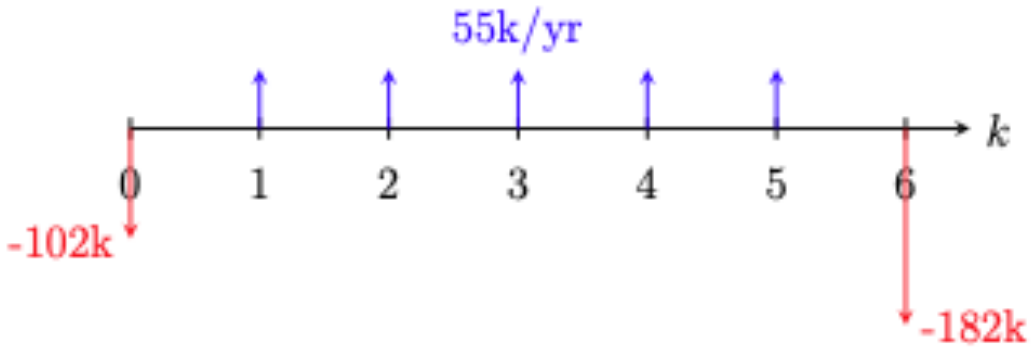
Norstrom's Criterion

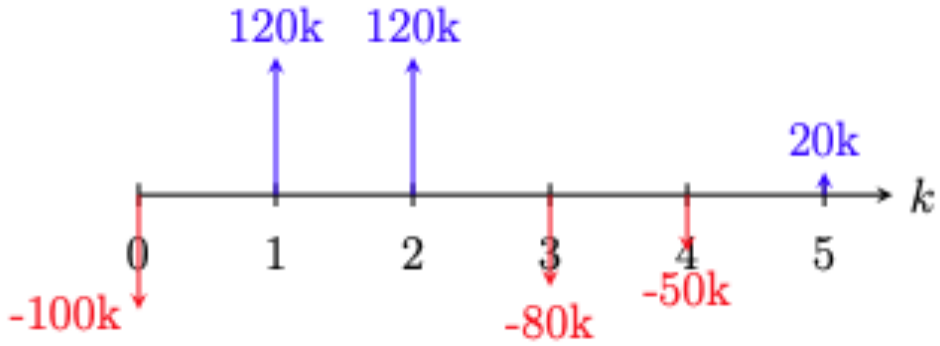
One sign change in a series of **cumulative** cash flow + **from negative to positive**



A unique **positive** ROR

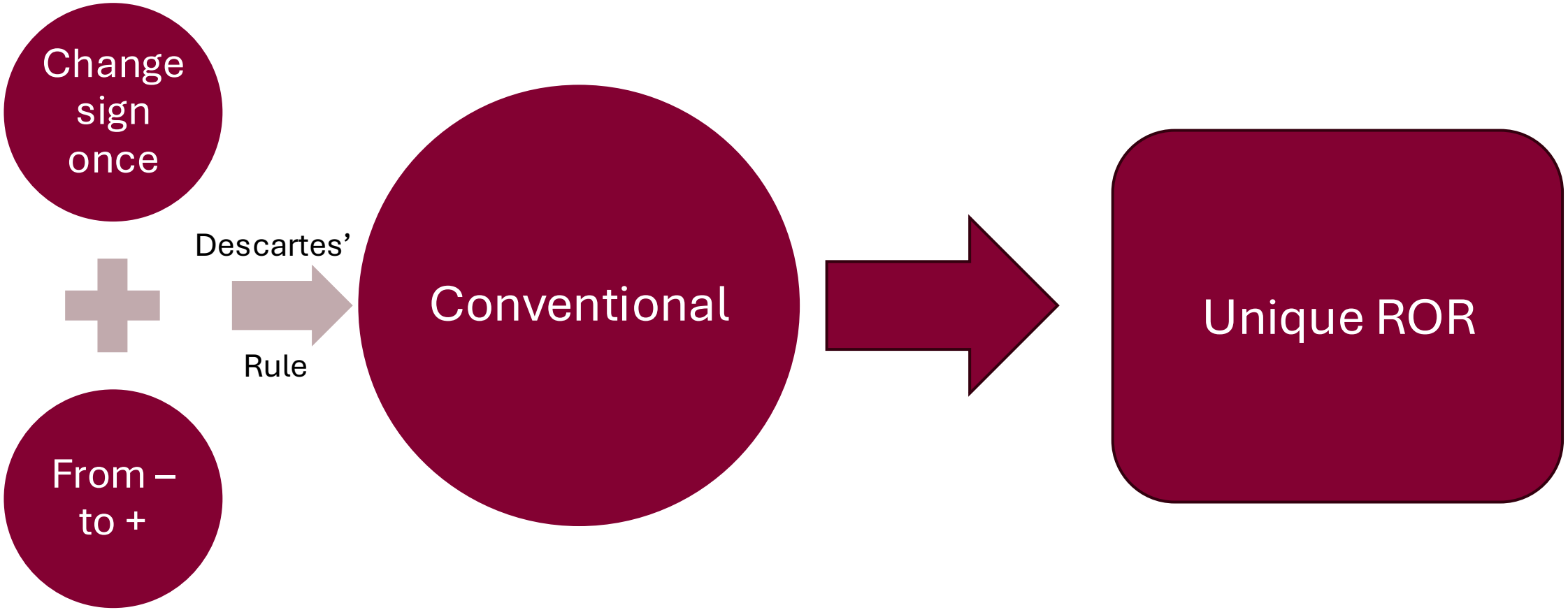
Identifying Multiple RoR - Examples



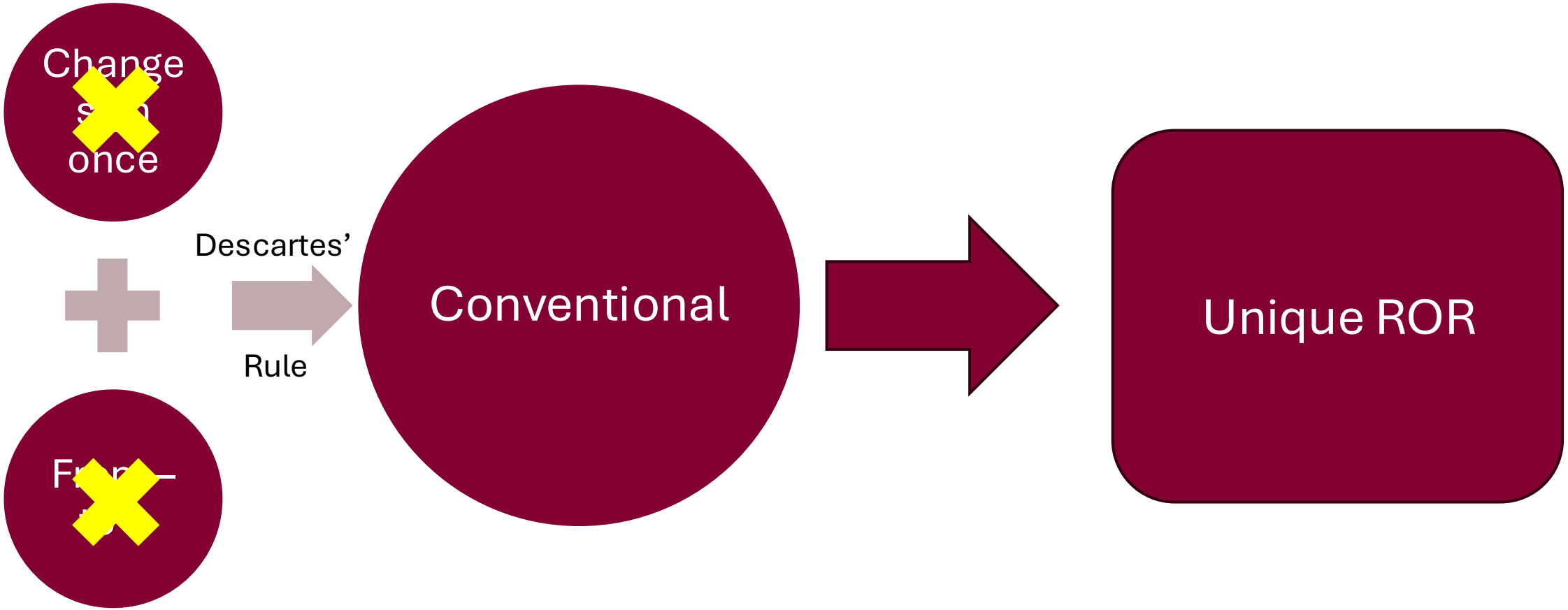


Identify the number of ROR Values for these two cash flows using both methods

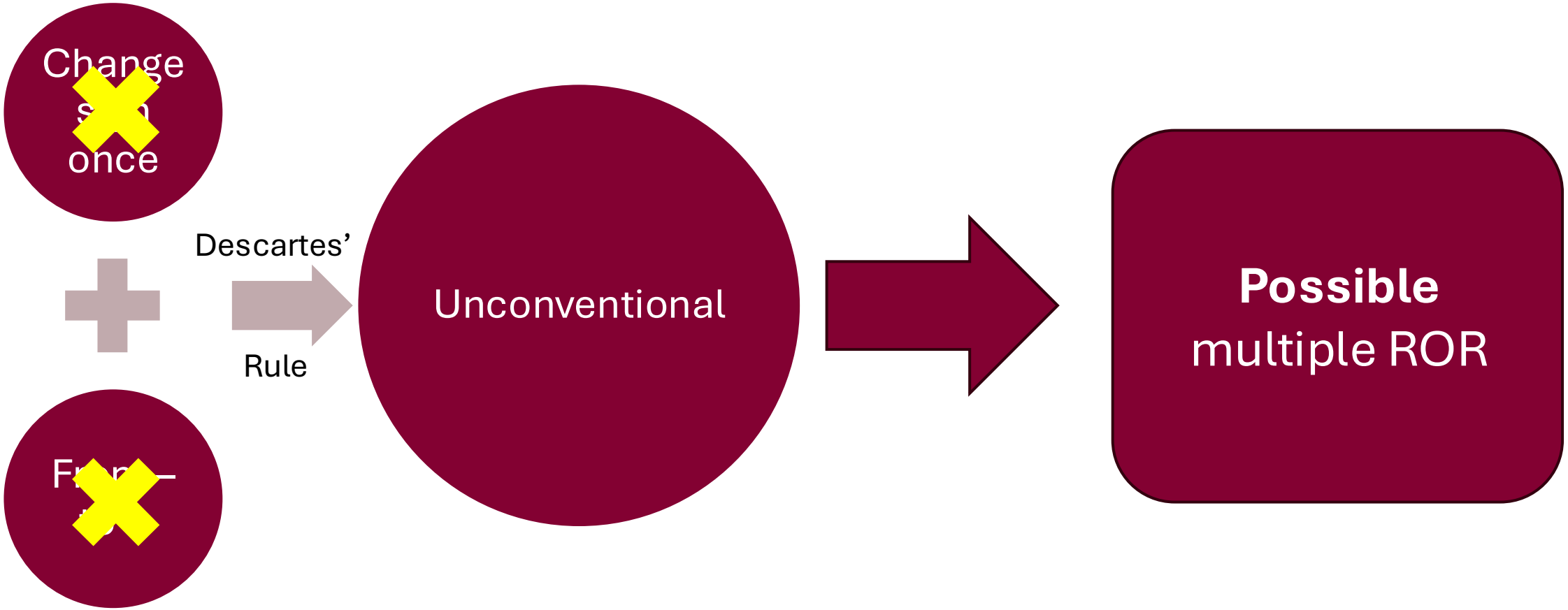
Conventional/Unconventional Cash Flow



Conventional/Unconventional Cash Flow



Conventional/Unconventional Cash Flow



Guidelines for Dealing with Multiple ROR

The ROR Values are	Action
Both $i^* < 0$	Discard both
Both $i^* > 0$	Discard both
One $i^* < 0$ & One $i^* > 0$	Use positive i^* as ROR

Assuming two ROR Values



ROR for Single Alternative- Summary

- The interest rate that sets the present worth of all cash flows to zero: $PW(i^*) = 0$
- If
  - $i^* \geq MARR$ → **Accept** alternative
  - $i^* < MARR$ → **Reject** alternative
- Some cashflows have multiple ROR values
  - Descartes' Rule
  - Norstrom's criterion

How to compare multiple alternatives?

